

Question ID: c602140f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question
 $(x - 11y)(2x - 3y) - 12y(-2x + 3y)$

Which of the following is equivalent to the expression above?

- Answer**
- A. $x - 23y$
- B. $2x^2 - xy - 3y^2$
- C. $2x^2 + 24xy + 36y^2$
- D. $2x^2 - 49xy + 69y^2$

Correct Answer: B

Rationale

Choice B is correct. Expanding all terms yields $(x - 11y)(2x - 3y) - 12y(-2x + 3y)$, which is equivalent to $2x^2 - 22xy - 3xy + 33y^2 + 24xy - 36y^2$. Combining like terms gives $2x^2 - xy - 3y^2$.

Choice A is incorrect and may be the result of using the sums of the coefficients of the existing x and y terms as the coefficients of the x and y terms in the new expressions. Choice C is incorrect and may be the result of incorrectly combining like terms. Choice D is incorrect and may be the result of using the incorrect sign in front of the 12y term.